

Aluminum 7072-H113

Categories: [Metal](#); [Nonferrous Metal](#); [Aluminum Alloy](#); [7000 Series Aluminum Alloy](#)

Material Notes: Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.

Composition Notes:

Composition information provided by the Aluminum Association and is not for design.

Key Words: UNS A97072; ISO AlZn1; Aluminium 7072-H113; AA7072-H113; Al7072-H113

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	2.72 g/cc	0.0983 lb/in ³	AA; Typical
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	21	21	500 kg load with 10 mm ball. Calculated value.
Tensile Strength, Ultimate	75.0 MPa	10900 psi	
Elongation at Break	15 % @Thickness 1.60 mm	15 % @Thickness 0.0630 in	In 5 cm
Tensile Modulus	68.0 GPa	9860 ksi	
Compressive Modulus	70.0 GPa	10200 ksi	
Poissons Ratio	0.33	0.33	Estimated from trends in similar Al alloys.
Shear Modulus	26.0 GPa	3770 ksi	Estimated from similar Al alloys.
Shear Strength	50.0 MPa	7250 psi	Calculated value.
Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.00000290 ohm-cm	0.00000290 ohm-cm	
Thermal Properties	Metric	English	Comments
CTE, linear 	21.8 µm/m-°C @Temperature -50.0 - 20.0 °C	12.1 µin/in-°F @Temperature -58.0 - 68.0 °F	
	23.6 µm/m-°C @Temperature 20.0 - 100 °C	13.1 µin/in-°F @Temperature 68.0 - 212 °F	AA; Typical; average over range
	24.5 µm/m-°C @Temperature 20.0 - 200 °C	13.6 µin/in-°F @Temperature 68.0 - 392 °F	
	25.5 µm/m-°C @Temperature 20.0 - 300 °C	14.2 µin/in-°F @Temperature 68.0 - 572 °F	
Specific Heat Capacity	0.893 J/g-°C	0.213 BTU/lb-°F	
Thermal Conductivity	222 W/m-K	1540 BTU-in/hr-ft ² -°F	
Melting Point	640.6 - 657.2 °C	1185 - 1215 °F	AA; Typical range based on typical composition for wrought products >= 1/4 in. thickness
Solidus	640.6 °C	1185 °F	AA; Typical
Liquidus	657.2 °C	1215 °F	AA; Typical

Processing Properties	Metric	English	Comments
Annealing Temperature	343 °C	650 °F	

Component Elements Properties	Metric	English	Comments
Aluminum, Al	<= 97.5 %	<= 97.5 %	As remainder
Copper, Cu	<= 0.10 %	<= 0.10 %	
Magnesium, Mg	<= 0.10 %	<= 0.10 %	
Manganese, Mn	<= 0.10 %	<= 0.10 %	
Other, each	<= 0.05 %	<= 0.05 %	
Other, total	<= 0.15 %	<= 0.15 %	
Si+Fe	<= 0.70 %	<= 0.70 %	
Zinc, Zn	0.80 - 1.3 %	0.80 - 1.3 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.